Local Execution Guide

Blue Harvest · Mortgage Pricing Exceptions

1. Prerequisites and Environment Configuration

Use a complete repository checkout and Windows PowerShell. The repository root contains compose.yaml, Backend and FrontEnd. Docker Desktop must use Linux containers.

Docker execution requires Docker Desktop, Docker Compose and a browser. Host execution and tests also require .NET 10 SDK and Node.js 24 with npm. Browser tests use Microsoft Edge on Windows. Initial downloads require internet access.

Defaults: frontend 5173, API 5267 and PostgreSQL 5432. Compose sets Development and the database connection. Optional root .env overrides: FRONTEND_PORT, BACKEND_PORT, POSTGRES_PORT and POSTGRES_PASSWORD. Defaults use database pricing_exceptions, user pricing and password local-demo-password. Changing the variable does not rotate a password in an existing database volume.

2. Commands to Start the Backend and Frontend

Use this host-development option instead of the full Docker stack. At the repository root, start PostgreSQL:

```
docker compose stop
docker compose up -d postgres
```

Terminal 1, repository root: start the backend. The HTTP profile sets Development and port 5267.

```
dotnet run --project Backend/BlueHarvest.PricingExceptions/src/PricingExceptions.API --launch-profile http
```

Terminal 2, repository root: start the frontend on port 5173.

```
cd FrontEnd
npm.cmd ci
npm.cmd run dev -- --host localhost --port 5173 --strictPort
```

Vite forwards API calls to localhost:5267. For custom host settings, set API_PROXY_TARGET for Vite and ConnectionStrings__PricingExceptions for the API. Stop host processes with Ctrl+C.

3. Docker Execution

Recommended: run the frontend, API and database together. From the repository root, run:

```
docker compose up -d --build
docker compose ps
```

Wait for PostgreSQL to become healthy and the API to finish starting. Open http://localhost:5173 for the frontend or http://localhost:5267/swagger for Swagger. Do not run host and Docker application processes on the same ports.

To stop the Docker application, run from the repository root:

```
docker compose stop
```

The named database volume retains data. Do not use down -v unless you intend to delete the database.

4. Database Initialization and Seed Data

On a new volume, PostgreSQL creates the configured database and database user. The API applies checked-in EF Core migrations automatically before serving requests. No manual schema creation is required.

In Development, startup creates or updates nine stable demo identities: three managers, five reviewers and one mortgage-process user. Existing requests and historical attribution are retained. Mortgage applications and example exception requests are not seeded; use a new non-empty application GUID.

Select users in the frontend. Direct Development API calls use X-User-Id for a seeded identity. The API reads the stored role. This simulates identity; it is not production authentication and demo headers are rejected outside Development.

5. Verify That Another Engineer Can Run the Solution Locally

5.1 In the frontend, select James Wilson (Manager). Create a request with a new mortgage application GUID, a 50-basis-point discount and a reason. Assign David Miller.

5.2 Switch to David Miller and approve. Switch to Demo Mortgage Process and look up the application GUID; the discount should be 50 basis points.

5.3 Switch back to James and submit revised terms of 75 basis points. The request becomes Pending. Discount lookup returns 404 until the new revision is approved. Open history and confirm both revisions and the earlier decision remain visible.

5.4 Backend tests: from a new terminal at the repository root, with Docker running:

```
cd Backend/BlueHarvest.PricingExceptions
dotnet build
dotnet test
```

5.5 Frontend checks: from another terminal at the repository root:

```
cd FrontEnd
npm.cmd ci
npm.cmd run build
npm.cmd run lint
npm.cmd run test:e2e
```

Leave E2E_BASE_URL unset. Isolated browser tests start servers on ports 5174 and 5270.

5.6 API smoke test: with the full Docker application running, execute from the repository root:

```
powershell.exe -NoProfile -ExecutionPolicy Bypass `
-File Backend/BlueHarvest.PricingExceptions/tests/Invoke-ApiSmokeTests.ps1 `
-BaseUrl http://localhost:5173
```

The smoke test creates demo records. Investigate failed commands or nonzero exit codes. Commands match repository configuration; tests were not rerun for this document update.